MIL-M-28787/223A(EC)
AMENDMENT 2
JUL 6 1983
SUPERSEDING
AMENDMENT 1
18 February 1982

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE KDC

This amendment forms a part of Military Specification MIL-M-28787/223A(EC), dated 12 March 1980, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2, second line: Delete "100°C" and substitute "85°C".

PAGE 2

* 1.2, Propagation delay time, t_{PLH} and t_{PHL}: Delete in its entirety.

PAGES 6, 7, and 8

* TABLE II, Delete and substitute new table II as printed on pages 3, 4, and 5 of this amendment.

PAGE 9

4.3, delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGE 10

* 6.3, delete and substitute the following:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/223 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833695. Part number M28787/223-1 supersedes part numbers 2833695 and M28787/223."

PAGE 14

TABLE III, Test conditions A1 and A2, Max column: Add "∞" (two places).

PAGE 16

* FIGURE 6: Delete and substitute new figure 6 as printed on page 6 of this amendment.

PAGE 24

* FIGURE 9: Delete and substitute new figure 9 as printed on page 7 of this amendment.

PAGE 25

TABLE VIII, Test condition F4, Parameter column: Delete "D1 = L" and substitute "D1 = H".

* TABLE VIII, Note 2, first line: Delete "V03" and substitute "M02".

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PAGE 26

* FIGURE 10: Delete and substitute new figure 10 as printed on page 8 of this amendment.

PAGE 29

* FIGURE 11: Delete and substitute new figure 11 as printed on page 9 of this amendment.

PAGE 38

TABLE XIV, Test condition M1, D to pin column: Delete "24" and substitute "25".

PAGE 39

TABLE XIV, Test condition M20, D to pin column: Delete "40" and substitute "18".

PAGE 57

TABLE XV, Test condition V20, B to pin column: Delete "30" and substitute "39".

PAGE 65

FIGURE 24, NOTE 3: Delete "120 Hz" and substitute "1 kHz".

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N503)

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B10	V_{OUT} (low-loaded)	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1-C26	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D10	Output short circuit current	VI	X	-20	-120	-20	-120	-20	-120	-20	-120	-20	-120	mA
E1-E26	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F5	Functional	VIII	X			See table VIII								
G1-G10	Static output voltage, low level	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H10	Static output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J26	Input current, high level	XI	X	0	40	0	40	0	40	0	40	0	40	µA
K1-K26	Input current, low level	XII	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA
L	Power supply current	XIII	S	10	150	10	150	10	150	10	150	10	150	mA
M1-M10	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	4	30	4	30	4	35	4	30	4	40	ns
M11-M20	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	4	40	4	40	4	45	4	40	4	50	ns
N1-N10	Transition time (t_{THL}), C_L = 50 pF	XIV	S	4	20	4	20	4	25	4	20	4	30	ns
N11-N20	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	4	25	4	25	4	30	4	25	4	35	ns

- - -

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P20	Propagation delay time (t_{PLH}) data to output, C_L = 50 pF	XIV	S	3	31	3	31	3	35	3	31	3	39	ns
P21-P40	Propagation delay time (t_{PLH}) data to output, C_L = 150 pF	XIV	Q	3	41	3	41	3	45	3	41	3	49	ns
Q1-Q10	Propagation delay time (t_{PLH}) select to output, C_L = 50 pF	XIV	S	6	40	6	40	6	49	6	40	6	55	ns
Q11-Q20	Propagation delay time (t_{PLH}) select to output, C_L = 150 pF	XIV	Q	6	50	6	50	6	59	6	50	6	65	ns
R1-R10	Propagation delay time (t_{PLH}) enable to output, C_L = 50 pF	XIV	S	3	29	3	29	3	33	3	29	3	37	ns
R11-R20	Propagation delay time (t_{PLH}) enable to output, C_L = 150 pF	XIV	Q	3	39	3	39	3	43	3	39	3	47	ns
S1-S20	Propagation delay time (t_{PHL}) data to output, C_L = 50 pF	XIV	S	3	23	3	23	3	25	3	23	3	27	ns
S21-S40	Propagation delay time (t_{PHL}) data to output, C_L = 150 pF	XIV	Q	3	33	3	33	3	35	3	33	3	37	ns
T1-T10	Propagation delay time (t_{PHL}) select to output, C_L = 50 pF	XIV	S	6	39	6	39	6	41	6	39	6	43	ns
T11-T20	Propagation delay time (t_{PHL})select to output, C_L = 150 pF	XIV	Q	6	49	6	49	6	51	6	49	6	53	ns

See footnotes at end of table.

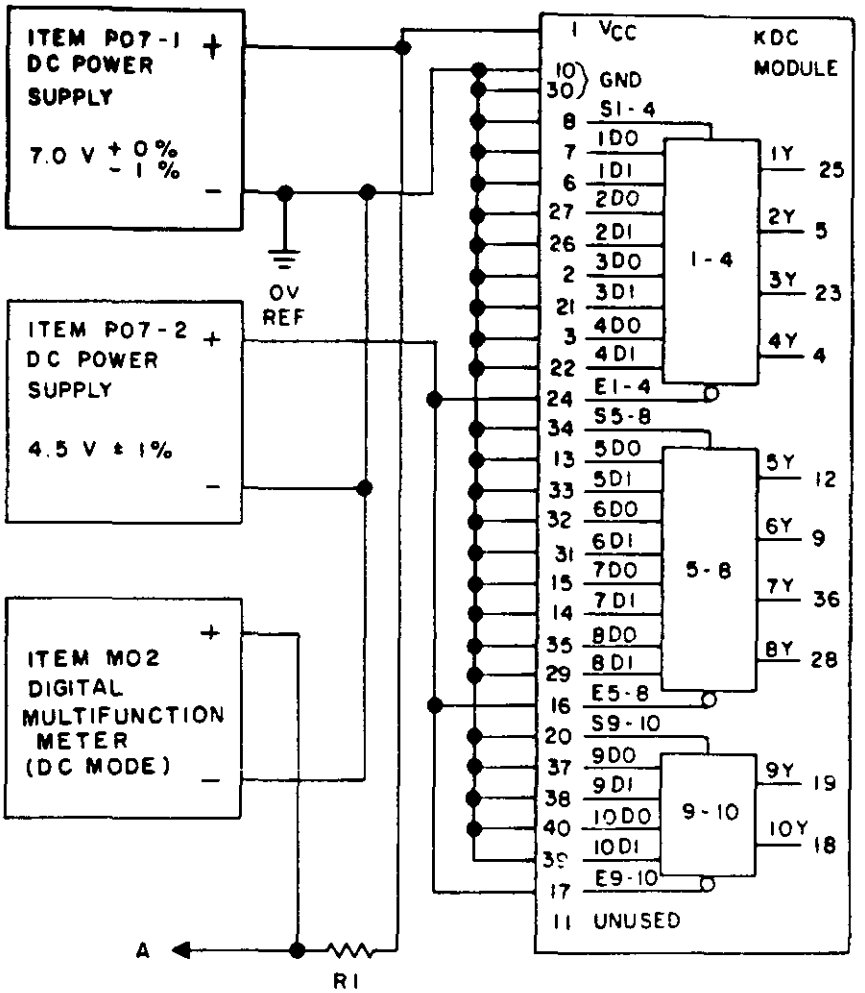
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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U10	Propagation delay time (t_{PHL}) enable to output, C_L = 50 pF	XIV	S	3	34	3	34	3	39	3	34	3	42	ns
U11-U20	Propagation delay time (t_{PHL}) enable to output, C_L = 150 pF	XIV	Q	3	44	3	44	3	49	3	44	3	52	ns
V1-V10	Output crosstalk, low level	XV	Q	0	1	0	1	0	1	0	1	0	1	V
V11-V20	Output crosstalk, low level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
W1-W10	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
Z1-Z26	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	2	15	pF
AA1	Ground continuity	XIX	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
AA2	Ground continuity	XIX	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

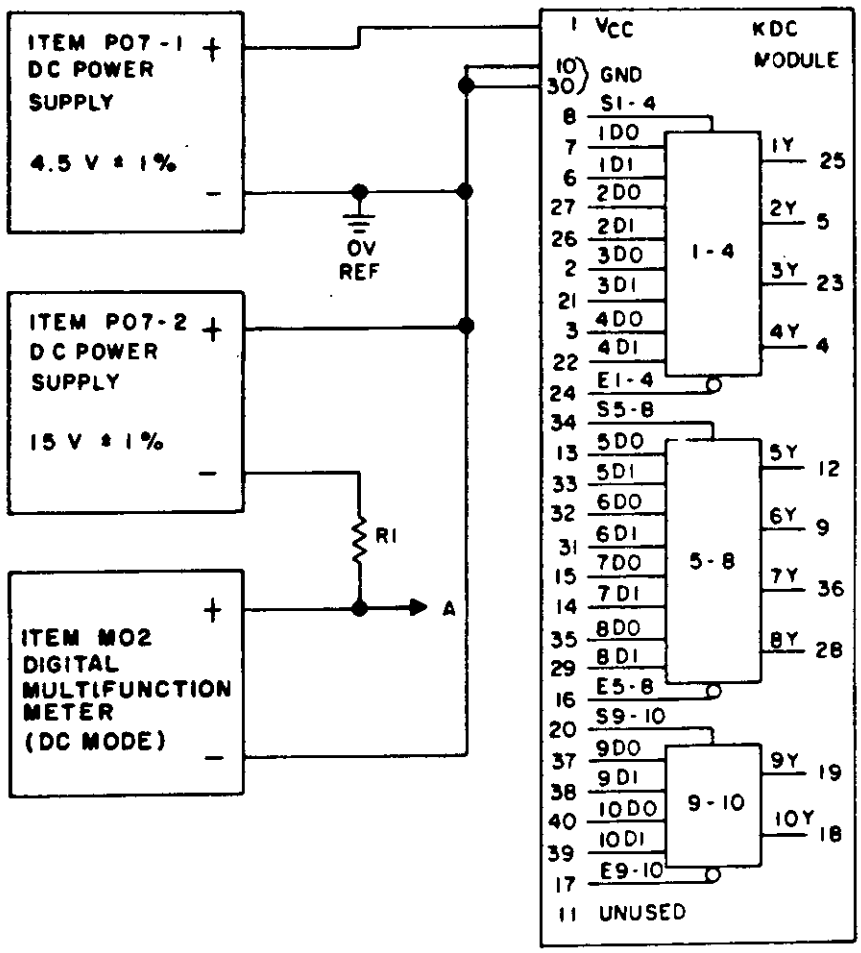
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NOTE: R1 = 324Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

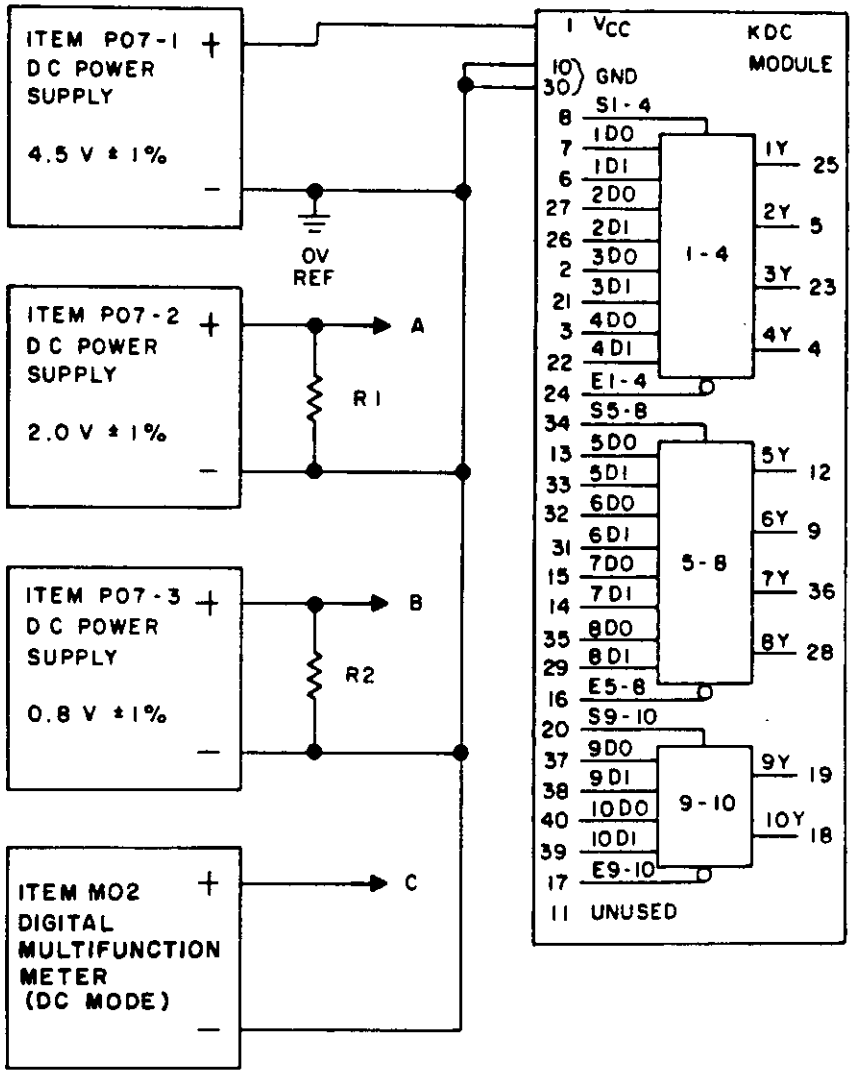
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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

FIGURE 9. Input clamp diode voltage test circuit for table VII.

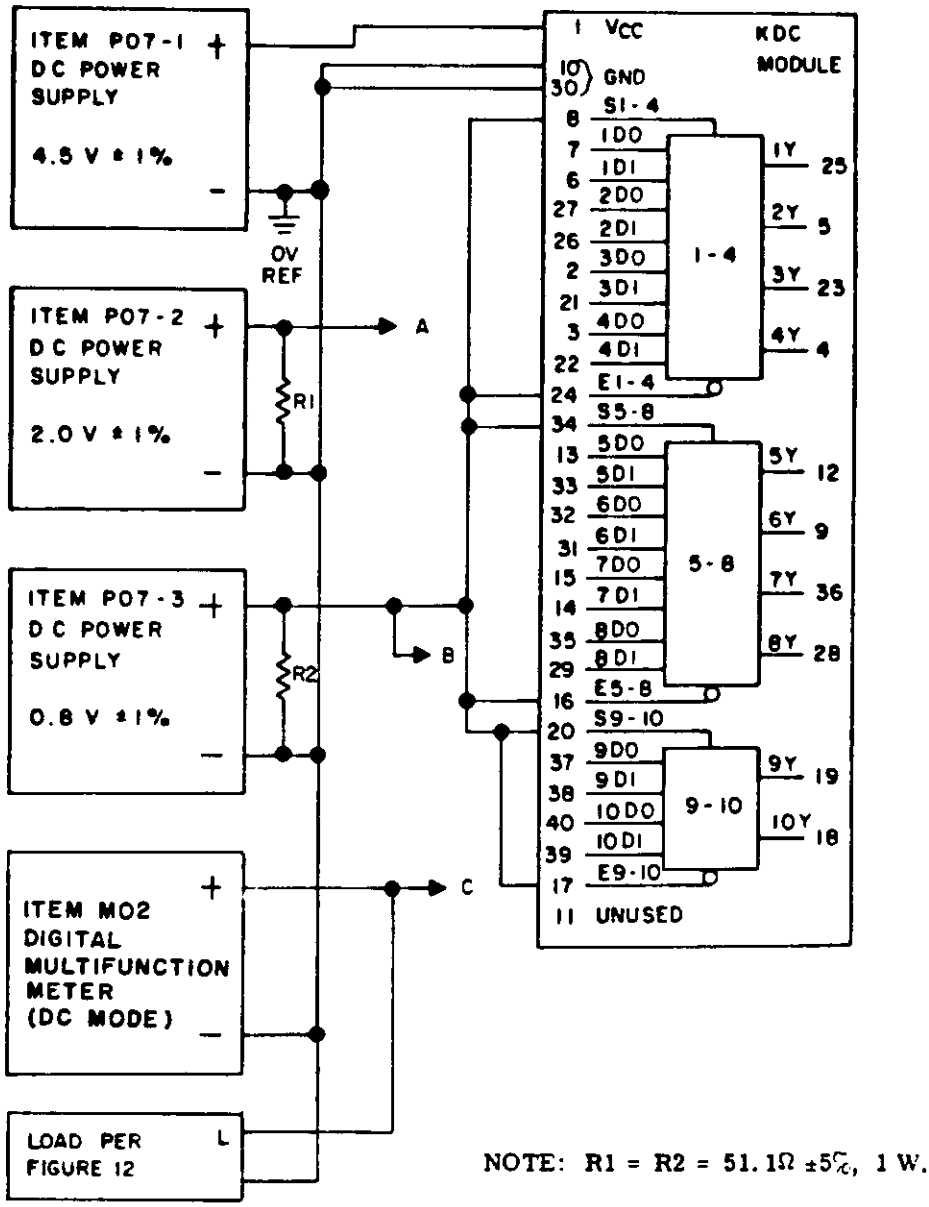
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NOTE: R1 = R2 = 51.1Ω ±5%, 1 W.

FIGURE 10. Functional test circuit for table VIII.

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NOTE: R1 = R2 = 51.1Ω ±5%, 1 W.

FIGURE 11. Static output voltage (low-level and high-level) test circuit for tables IX and X.

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